Process Characterization Report

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCR-007

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

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Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; CV, coefficient of variation; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PCMR, process characterization master report; PCP, process characterization plan; PCR, process characterization report; qPCR, quantitative polymerase chain reaction; RA, risk assessment; RSD, relative standard deviation; RSM, response-surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UV, ultraviolet; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Cation exchange chromatography is the first of the two polishing steps in the A-Mab drug substance process (Step 7), and it is the only step in the train that reduces aggregate. The step is operated in bind and elute mode on a strong cation exchange resin, with a step elution and a defined end of collection on the descending edge of the product peak. This report covers the characterization study executed under PCP-007 in a qualified scale-down model, on the risk basis established in RA-001. The study addressed 5 process parameters, of which 4 were taken into a multivariate design of 19 screening runs and 28 response-surface runs, while the remaining parameter (the elution flow rate) was assessed univariately in a separate block. The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11, and it takes the process description, the attribute set and the criticality framework from the A-Mab case study (CMC Biotech Working Group 2009).

The step sets no quality attribute of its own. It clears four of them: aggregate (high molecular weight species by size-exclusion chromatography), host cell protein (HCP), residual DNA and leached Protein A. Aggregate is the attribute the step carries, because no later unit operation in the train reduces it. In the nominal commercial-scale run the step took the pool from 1.91 % to 0.73 % high molecular weight species, a reduction of 2.6-fold, and the response-surface model built here predicts a pool aggregate no higher than 1.82 % anywhere inside the characterized region, against a drug substance limit of 5 % (Table 2.1). Aggregate stays within its limit at every corner of the ranges studied, so the operating region defined in Section 6 is bounded by those ranges and not by the aggregate limit.

Pool host cell protein is the response that moves most across the design, and the step is judged on the clearance factor it delivers and not on a drug substance release criterion. In the nominal commercial-scale run the step reduced host cell protein 78-fold, from 9,125 ng/mg in the Protein A pool to 117 ng/mg. That pool value is above the drug substance criterion of 100 ng/mg, which is the expected position for an intermediate, because the drug substance does not exist until anion exchange chromatography has run. Anion exchange clears a further factor of 6.2 to 18.9 ng/mg (PCR-008), and the cumulative position across the train is consolidated in PCMR-001. Pool host cell protein rises with protein load, falls as the conductivity of the load and wash buffer rises, and the two parameters act together, so the least favourable corner of the characterized region (high protein load with low conductivity) is the one the step must still satisfy. The model predicts 291 ng/mg there, against a step-level ceiling of 619 ng/mg derived in Section 6 from the anion exchange clearance factor.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches operated within their normal operating ranges (NOR). Of the four attributes the step governs, the tightest capability belongs to host cell protein, at a Cpk of 6.1 against a mean of 15.2 ng/mg. Aggregate is far wider, at a Cpk of 16.1 on a mean of 0.75 % high molecular weight species. Both figures are conditional on operation within the normal operating ranges and on the validity of the scale-down model, and neither is a statement about behaviour at the edges of the region defined in Section 6.

All 4 multivariate parameters are classified as well-controlled critical process parameters (WC-CPP), and the elution flow rate is a general process parameter (GPP). No parameter at this step required classification as a critical process parameter, and none is a key process parameter. The campaign recorded 2 deviations during execution, both were investigated and dispositioned as retained, and neither altered a fitted effect, an operating region boundary or a classification (Section 13). No viral clearance claim is made for this step, and no claim is made for modification of charge or glycan variants. The outcomes of the study roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized immunoglobulin G subclass 1 monoclonal antibody produced in a fed-batch mammalian cell culture. Its principal mechanism of action is antibody-dependent cellular cytotoxicity, so the glycan attributes that govern effector function (afucosylation, galactosylation and high mannose) are formed in the production bioreactor and are not altered downstream (CMC Biotech Working Group 2009). The drug substance process comprises eight steps, listed with their control-strategy roles in Table 1.1. Cation exchange chromatography is Step 7. It receives the neutralized pool from low-pH viral inactivation (Step 6, reported in PCR-006) and delivers the load for anion exchange chromatography (Step 8, reported in PCR-008).

Table 1.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The step uses a strong cation exchange resin in bind and elute mode. The load and wash buffer is held at a conductivity low enough for the antibody to bind, while weakly retained process impurities (host cell protein above all) pass into the flow-through and the wash. Product is recovered by a step elution, and collection ends at a defined criterion on the descending edge of the elution peak. Three mechanisms give the step its clearance. The first of them is the reason the step carries the aggregate claim, since aggregated species carry a higher surface charge density than monomer, bind more strongly, and are enriched in the descending edge of the peak, so that a pool cut placed before that tail removes most of them. Host cell protein species that bind through weaker electrostatic interactions are removed by the ionic strength of the load and the wash. Residual DNA and leached Protein A are removed by those same two mechanisms and are credited to the step at fixed nominal levels (Section 2.2).

The step is a polishing operation and forms no quality attribute. It is nonetheless the step on which the aggregate content of the drug substance depends, since none of the operations that follow it (anion exchange chromatography, small-virus retentive filtration and ultrafiltration) reduces aggregate. The characterization is built around the aggregate response for that reason, and the acceptance argument in Section 6 is made against the drug substance aggregate limit and not against an in-process limit.

1.2 Objectives

The study was executed to meet the objectives set out in PCP-007. They were:

- (i) Quantify the relationship between each process parameter and the measured responses over ranges wider than the normal operating range.
- (ii) Resolve the interactions between parameters that prior knowledge indicated were likely, in particular between protein load and the conductivity of the load and wash buffer.
- (iii) Define the multivariate operating region over which the attributes the step governs remain within their acceptance criteria.
- (iv) Establish a proven acceptable range for each parameter against each governed attribute, both at the set-point and with the remaining parameters varying within their normal operating ranges.
- (v) Classify each parameter and state the resulting contribution of the step to the control strategy for commercial manufacture.

1.3 Regulatory and scientific basis

The study is a Stage 1 activity under the lifecycle approach to process validation, in which Stage 1 is process design and Stage 2 is process qualification (U.S. Food and Drug Administration 2011). It provides the process understanding on which the Stage 2 ranges (the ranges qualified for the commercial step) and the commercial control strategy are built. The enhanced development approach of ICH Q8(R2) supplies the design space concept used in Section 6, and ICH Q11 supplies the link from process understanding to the control strategy (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum in the sense of ICH Q9(R1), so that quality attributes carry a graded criticality (from very low to very high) and each process parameter carries a class that reflects both its demonstrated effect on those attributes and the reliability with which it can be held inside the operating region (International Council for Harmonisation 2023b). The process model, the attribute set and the criticality framework are those of the A-Mab case study (CMC Biotech Working Group 2009).

No viral clearance claim is made for this step. Cation exchange chromatography can contribute to enveloped-virus removal, but the modular clearance claims for A-Mab rest on low-pH inactivation, anion exchange chromatography and small-virus retentive filtration (Steps 6, 8 and 9), and they are evaluated under ICH Q5A(R2) in PCR-006, PCR-008 and PCR-009 (International Council for Harmonisation 2023a). The documents that frame, scope and consolidate this report are listed in Table 1.2.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-007	Process Characterization Plan (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The cation exchange polishing step is a platform operation. It has been operated at commercial scale for three earlier humanized immunoglobulin G subclass 1 products (X-Mab, Y-Mab and Z-Mab) and through the A-Mab clinical campaigns, on the same resin chemistry and the same buffer system. The accumulated experience shows that the step reduces aggregate and host cell protein reproducibly, and that its performance is governed by the loading and the elution conditions and not by the resin lot. Characterization was therefore designed to confirm and bound the platform behaviour for A-Mab, and not to discover it.

Prior knowledge also fixed the direction expected of each parameter, which is what makes the study a test and not a survey. Protein load was expected to raise both pool aggregate and pool host cell protein, because a column operated closer to its capacity resolves the strongly bound aggregate and the weakly bound impurity less completely, and it was also expected to lower step yield as product began to appear in the flow-through. The conductivity of the load and wash buffer was expected to reduce pool host cell protein, since raising the ionic strength disrupts the weaker electrostatic interactions that retain impurity while leaving the antibody bound. Elution buffer pH and the elution stop collect criterion were expected to act on aggregate through the position and the width of the collected peak, and to act on host cell protein very little, because host cell protein is removed before the elution begins. The elution flow rate was expected to act on cycle time alone. These expectations are examined against the data in Section 5.4.

2.2 Quality attributes in scope

This step forms no quality attribute, so the register of attributes it sets is empty. The attributes it controls are the four impurity and size-variant attributes listed in Table 2.1, each of which enters the step from an upstream operation and leaves it reduced. Aggregate carries a high criticality and is measured as high molecular weight species by size-exclusion chromatography (AMV-3011). Host cell protein carries a moderate to high criticality and is measured by immunoassay (AMV-3012). Residual DNA and leached Protein A carry low criticality and large capability margins, and they are reported here for completeness of the clearance account rather than because this step governs them.

Table 2.1: Quality attributes controlled and cleared by the cation exchange step, with their drug substance acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Two of the four attributes were measured as responses across the designed experiments. Aggregate and host cell protein vary with the operating conditions over the ranges studied and were modelled. Residual DNA and leached Protein A are removed to a nominal 1.5 and 1.0 log reduction respectively at this step, and neither was carried as a design response, because the clearance credited to the step is small beside the margin those attributes hold at the drug substance stage (see Table 8.1). Their commercial-scale capability is given in Section 8, and the steps that carry the larger part of their clearance are named there.

Aggregate is the attribute for which this step is the last opportunity. No further reduction or clearance step for aggregate exists downstream of cation exchange, so whatever aggregate leaves this pool is the aggregate content of the drug substance, and the acceptance argument for the attribute has to be made here in full rather than shared with a later operation. Host cell protein, residual DNA and leached Protein A are treated differently, because anion exchange chromatography reduces all three again (PCR-008), and Protein A chromatography carries the largest single share of the host cell protein and DNA reduction (PCR-005).

2.3 Risk-based prioritization and parameter selection

The scope of this study was fixed by the pre-characterization risk assessment RA-001, which ranked every parameter of the step on its potential effect on the governed attributes and on its potential to interact with other parameters. Parameters that scored high on either axis were assigned to a multivariate study, and parameters that scored low on both were assigned to a univariate assessment. 4 parameters met the multivariate criterion: protein load, the conductivity of the load and wash buffer, elution buffer pH and the elution stop collect criterion. The elution flow rate did not, because prior knowledge places its action on residence time and on cycle time, and not on the binding chemistry that governs the separation.

The ranges taken into the study are wider than the ranges the step will be operated within. Each characterization range spans the normal operating range and extends beyond both of its edges, which is what allows the study to describe process behaviour at conditions the commercial process will not routinely see, and to support later movement within the operating region. The ranges and their relationship to the set-points and the normal operating ranges are given in Table 4.1. Two of them were set by specific prior constraints. The protein load range was extended upward to a value well above the routine load (its upper edge is 40 g/L resin against a set-point of 25 g/L resin), so that the approach to the dynamic binding capacity of the resin would be visible in both the aggregate and the yield responses. The elution pH range was kept narrow, because the elution buffer is prepared to a tight specification and a wider excursion would move the elution position far enough to make the collected peak unrepresentative of the commercial operation.

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were performed in a qualified scale-down model of the commercial cation exchange step, operated under SOP-2010 and qualified under SOP-1001. The model preserves the parameters that determine chromatographic behaviour. Bed height, linear flow velocity, protein load per unit resin volume, and the load, wash and elution volumes expressed in column volumes are all held at their commercial values, so that the residence time and the number of column volumes seen by the resin match at both scales. Column packing quality was confirmed before each block by plate count and peak asymmetry under SOP-2008. Buffer systems, resin lot handling and the sanitization regime were the commercial ones, so that the only deliberate differences between the model and the commercial step are the column diameter and the absolute volumes handled.

Qualification of the model rests on the agreement of its output attributes with the commercial-scale process at equivalent operating conditions. At the centre point the model returns a pool aggregate of 0.75 % high molecular weight species against 0.73 % for the nominal commercial-scale run, a relative difference of 2.6 %. The corresponding pool host cell protein values are 141 ng/mg and 117 ng/mg, which differ by 1.3 standard deviations of the centre-point replicate spread of that response (Table 5.1). The aggregate agreement is the material one, because aggregate is the attribute the step carries and no later step can correct it, and the host cell protein difference is smaller than the reproducibility of the immunoassay used to measure it. The model is accordingly regarded as representative of the commercial step for the purpose of defining operating ranges. It is not used to support any viral clearance claim, and the qualification record is held with the study data referenced in PCP-007.

3.2 Operation

Each run was executed as a complete chromatographic cycle. The column was equilibrated in the load and wash buffer at the conductivity assigned to the run, loaded to the assigned protein load, washed with the same buffer, and eluted with the elution buffer at the assigned pH. Collection of the pool began at a fixed criterion on the ascending edge of the elution peak and ended at the assigned stop collect criterion on the descending edge. Between runs the column was cleaned and sanitized under SOP-2008. Inputs that are controlled to a platform specification (the resin, the buffer components and the water quality) were verified against those specifications and were not varied as study factors, because they are identity and quality controlled and carry no residual uncertainty that this study could reduce.

The load material for every run was cation exchange load derived from a single low-pH viral inactivation pool, so that the impurity and aggregate content of the feed was held constant across the design. Holding the feed constant is what allows the fitted effects to be attributed to the operating parameters. It also bounds the study, since the effect of a feed with a different aggregate or host cell protein content is outside the region the models describe. That bound is stated again in Section 11.

3.3 Analytical methods

Every response was measured by a validated method. Pool aggregate was measured as high molecular weight species by size-exclusion chromatography (SEC-HPLC) under AMV-3011, host cell protein by immunoassay (ELISA) under AMV-3012, residual DNA by quantitative polymerase chain reaction (qPCR) under AMV-3014 and leached Protein A by immunoassay under AMV-3016. Step yield was calculated from the product mass in the pool and the product mass in the load, each determined by ultraviolet absorbance (A280). The controlled documents governing the step and its methods are listed in Table 3.1, and method performance characteristics are summarized in Appendix D.

Table 3.1: Controlled documents and validated analytical methods applied to the cation exchange characterization.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

The host cell protein immunoassay is the least precise of the methods used here (a relative standard deviation of 9.5 %), and its contribution to the observed variation is quantified in Section 5.1. That is a property of the method and not of the process, and it is the reason the host cell protein model is interpreted through its fitted effects and not through individual run values.

3.4 Sampling plan

One pool sample was taken from every run and assayed for aggregate, host cell protein, residual DNA and leached Protein A. Load samples were taken from each load preparation and assayed for the same four attributes, so that clearance across the step could be calculated run by run and not only as a difference of block means. Flow-through and wash fractions were retained but were not assayed routinely. The screening block carried 3 centre-point runs and the response-surface block carried 4, distributed through the run order so that the replicate spread reflects the full execution period (and therefore any slow drift in the column or the assays) and not a single session.

3.5 Statistical methods

All models were fitted on coded factors. Each factor was scaled so that its characterization range maps to the interval from minus one to plus one, with the set-point at zero (for all four multivariate factors the set-point is also the midpoint of the characterization range), which makes the estimated coefficients directly comparable across parameters with different units. The screening data were fitted with a model containing all main effects and all two-factor interactions, and effects are reported as twice the coded coefficient, so that an effect is the change in the response across the full characterization range of its factor. The response-surface data were fitted with the full quadratic model in the same four factors. Significance was assessed at the 5 % level throughout.

Model adequacy was judged on four grounds: the coefficient of determination and its adjusted form, the predicted coefficient of determination computed from the prediction error sum of squares, the overall F test, and an analysis of variance that partitions the residual into lack of fit and pure error. The pure-error term comes from the replicated centre points, so the lack of fit test asks whether the model misses structure that the replicates say is real, and a model that passes it is one whose unexplained variation is no larger than the reproducibility of the step itself. A response for which no factor reaches significance is reported as robust over the ranges studied and is retained in the knowledge space on that basis.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with each parameter drawn within its normal operating range and the attribute computed from the process model. Capability is expressed as a one-sided Cpk against the relevant drug substance criterion, since all four attributes governed here are impurities with an upper limit only. Proven acceptable ranges were derived from the fitted response-surface models in two forms (at the set-point and with the other parameters varying within their normal operating ranges), and both are described in Section 7.

4 Study design

4.1 Factors, ranges and the knowledge space

The design covers 5 parameters, of which 4 were studied multivariately and 1 univariately. Table 4.1 gives the set-point, the normal operating range, the characterization range, the final classification and the study type for each. The characterization range is the knowledge space edge for that parameter. It is not in itself a proven acceptable range, which is a per-attribute quantity derived in Section 7.

Table 4.1: Cation exchange process parameters, with set-points, normal operating ranges, characterization ranges, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	25	18–32	10–40	WC-CPP	multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	multivariate
Elution stop collect	OD desc.	1	0.7–1.3	0.5–1.5	WC-CPP	multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	univariate

The four multivariate factors are coded A to D throughout the effect and coefficient tables. Table 4.2 gives the mapping and shows that all four were carried from the screening design into the response-surface design.

Table 4.2: Coded factor legend for the cation exchange designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

Each range was chosen against a stated constraint. The protein load range spans a wide band around the set-point (from 10 to 40 g/L resin), because load is the parameter with the largest expected effect and because the upper edge has to approach the working capacity of the resin for the study to describe an overloaded column. The conductivity range for the load and wash buffer is set by the two failure modes that bound it, since too low a conductivity retains impurity that the wash should remove, while too high a conductivity begins to compete with product binding and would cost yield. The elution pH range is narrow for the reason given in the risk section. The stop collect range spans a band on either side of the platform criterion (from 0.5 to 1.5) that is wide enough to move a visible amount of the descending edge into and out of the pool.

4.2 Screening design

The screening design was a two-level full factorial in the four multivariate factors, augmented with centre points. It comprises 16 factorial runs and 3 centre-point runs, for 19 runs in total, executed in randomized order. A full factorial in four factors estimates every main effect and every two-factor interaction without aliasing, which is what the risk assessment required, since the interaction between protein load and load and wash conductivity was expected in advance. The centre points provide the pure-error estimate and a test for curvature. The coded design matrix and the measured responses are given in Appendix A.

The purpose of this block is to identify which factors matter, and it is used for that purpose only. The model it supports contains eleven terms fitted to 19 runs, so the residual degrees of freedom are few, the fitted magnitudes are correspondingly uncertain, and the design is not able to separate curvature in one factor from curvature in another. The predictive model for this step is the response-surface model described next.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same four factors. It comprises 16 factorial runs, 8 axial runs placed on the faces of the design cube and 4 centre-point runs, for 28 runs in total. Placing the axial points on the faces keeps every run inside the characterization ranges, so that no run is executed at a condition the design does not claim to describe. The axial points supply the information needed to estimate the four quadratic terms and so to test whether any response curves over its range. The coded design matrix and the measured responses are given in Appendix B.

This design is the basis of every predictive statement in this report. The operating region in Section 6, the proven acceptable ranges in Section 7 and the worst-case predictions quoted in the executive summary all come from the fitted quadratic models, and not from the screening fit.

4.4 Univariate assessment

The elution flow rate was assessed univariately across its characterization range (from 100 to 300 cm/hr) with the four multivariate factors held at their set-points. Flow rate acts on residence time, and residence time affects the sharpness of the elution peak and not the binding chemistry that determines where impurity and aggregate partition. No effect on pool aggregate, pool host cell protein or step yield was detected across the range assessed, which is the basis for the classification given in Section 9. Keeping the parameter out of the multivariate design also kept that design small enough to execute in a single campaign, which is the practical reason risk-based prioritization is applied at all.

5 Results

5.1 Centre-point performance and reproducibility

The step reproduces well at its set-point, and the residual variation differs sharply between the three responses. Table 5.1 gives the mean, the standard deviation and the coefficient of variation of the 4 centre-point runs of the response-surface block.

Table 5.1: Centre-point reproducibility in the response-surface block, and the pure-error basis for the lack-of-fit tests.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	0.771	0.043	5.581
Pool HCP (ng/mg)	4	133.989	18.651	13.919
Step yield	4	0.918	0.029	3.152

Step yield is the most reproducible response, at a coefficient of variation of 3.2 %. Pool aggregate follows at 5.6 %. Pool host cell protein is the least precise at 13.9 %, which is close to the relative standard deviation of 9.5 % established for the immunoassay under AMV-3012, so most of the centre-point spread in that response is analytical variation and not process variation. The same pattern holds in the screening block, where the aggregate and host cell protein coefficients of variation were 10.2 % and 17.6 %. Both blocks are less precise for host cell protein than for the other two responses by the same factor, which is what a method limitation looks like and what a process instability would not. These replicate spreads are the pure-error terms used in the lack-of-fit tests reported in Section 5.3.

5.2 Screening: factor effects

The screening block identified protein load as the dominant parameter for every response, and resolved the two interactions that the risk assessment had anticipated. Table 5.2 summarizes the fit of the screening models.

Table 5.2: Screening model fit summary for the three measured responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	19	0.9727	0.9385	0.8691	28.49	3.568e-05	0.09599
Pool HCP (ng/mg)	19	0.9363	0.8566	0.4667	11.75	0.0009373	35.4
Step yield	19	0.9241	0.8292	0.4978	9.739	0.001811	0.01147

Pool aggregate is affected by protein load, elution buffer pH and the stop collect criterion, and by three interactions among them. The six significant terms are given in Table 5.3. Protein load has the largest effect, at 0.60 percentage points of high molecular weight species across its characterization range. Elution buffer pH and the stop collect criterion follow, at 0.34 and 0.32 points, and all three raise pool aggregate as they increase. The interaction between protein load and elution buffer pH is significant at 0.20 points, and the interactions of protein load and of elution pH with the stop collect criterion are significant and smaller (Table 5.3). The conductivity of the load and wash buffer has no detectable effect on pool aggregate at all, at -0.018 points and a p-value of 0.71, and that null result is carried forward as evidence that the aggregate and host cell protein controls act through different parameters.

Table 5.3: Screening effect estimates for pool aggregate, the six largest by absolute magnitude. Effects are expressed across the full characterization range of each factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.5987	0.2993	0.024	12.47	1.595e-06	***
C	0.3401	0.1701	0.024	7.087	0.0001033	***
D	0.319	0.1595	0.024	6.648	0.0001612	***
AC	0.1958	0.0979	0.024	4.08	0.003536	**
AD	0.1544	0.07718	0.024	3.216	0.0123	*
CD	0.1313	0.06565	0.024	2.736	0.02561	*

Pool host cell protein is affected by the conductivity of the load and wash buffer, by protein load, and by the interaction between them (Table 5.4). Conductivity has the largest effect, reducing pool host cell protein by 127 ng/mg across its range. Protein load raises it by 116 ng/mg across its range, which is a comparable magnitude in the opposite direction. The interaction is significant at -57 ng/mg, and its negative sign means that raising the conductivity removes more host cell protein when the column is loaded heavily than when it is loaded lightly, which is the effect the risk assessment predicted and the reason these two parameters were taken into a multivariate design together. Neither elution buffer pH nor the stop collect criterion has a significant effect on pool host cell protein (p-values of 0.61 and 0.18), which is consistent with a separation that takes place during the load and the wash and not during the elution.

Table 5.4: Screening effect estimates for pool host cell protein, the six largest by absolute magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-126.6	-63.29	8.85	-7.151	9.701e-05	***
A	116.4	58.22	8.85	6.579	0.0001732	***
AB	-56.75	-28.37	8.85	-3.206	0.0125	*
BD	39.6	19.8	8.85	2.237	0.05568	
D	-25.84	-12.92	8.85	-1.46	0.1824	
AD	-24.59	-12.3	8.85	-1.389	0.2022	

Step yield depends on protein load alone. The effect is -0.052 across the load range, and no other term reaches significance. That is a null result for three of the four factors, and it is retained as such, because the conductivity of the load and wash buffer, the elution pH and the stop collect criterion may all be moved across their whole characterization ranges without a measurable yield penalty. Their inactivity is evidence of robustness and is carried into the classification in Section 9.

The screening block is used to identify the active factors and nothing more. Its models fit eleven terms to 19 runs, and the predicted coefficient of determination for pool host cell protein is only 0.47, against 0.94 for the raw coefficient of determination (Table 5.2). The gap between those two figures is what a near-saturated design produces, and it is the reason no operating range and no proven acceptable range is drawn from this block. The effect magnitudes are refined by the response-surface models below, and it is those models that carry every predictive claim in this report.

5.3 Response-surface models

The response-surface models describe pool aggregate and pool host cell protein well, and step yield only in part. Table 5.5 gives the fit summary for all three.

Table 5.5: Response-surface model fit summary.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	28	0.9754	0.949	0.8234	36.86	3.851e-08	0.07486
Pool HCP (ng/mg)	28	0.9659	0.9292	0.827	26.32	3.067e-07	16.8
Step yield	28	0.8251	0.6367	0.196	4.379	0.005727	0.01883

Pool aggregate is described by a model with a coefficient of determination of 0.975, an adjusted value of 0.949 and a predicted value of 0.823. Pool host cell protein is described almost as well, at 0.966 and a predicted value of 0.827. The closeness of the predicted value to the adjusted value in both models is the property that matters for a predictive claim, because it says the models keep their accuracy on data they have not seen, and both models are used for prediction in this report on that basis. The step yield model reaches significance overall, at a p-value of 0.0057, but its predicted coefficient of determination is only 0.20. The yield model is therefore used to state the direction and the approximate size of the load effect, and it is not used to predict yield at any given condition. Yield is a process performance response and carries no quality claim, so that limitation does not touch the acceptance argument in Section 6.

Table 5.6: Full quadratic coefficients for pool aggregate, in model order. Coefficients are on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.7507	0.02589	29	3.356e-13	***
A	0.2804	0.01765	15.89	6.768e-10	***
B	0.02762	0.01765	1.565	0.1415	
C	0.172	0.01765	9.75	2.412e-07	***
D	0.1566	0.01765	8.875	7.05e-07	***
AB	0.009639	0.01872	0.515	0.6152	
AC	0.09604	0.01872	5.131	0.0001929	***
AD	0.08505	0.01872	4.544	0.0005507	***
BC	-0.002111	0.01872	-0.1128	0.9119	
BD	0.01372	0.01872	0.733	0.4766	
CD	0.06943	0.01872	3.71	0.002621	**
A^2	0.07976	0.04661	1.711	0.1108	
B^2	-0.008754	0.04661	-0.1878	0.8539	
C^2	0.02639	0.04661	0.5661	0.581	
D^2	0.0677	0.04661	1.452	0.1701	

The aggregate coefficients in Table 5.6 confirm the screening picture and sharpen it. Protein load carries a coefficient of 0.280, elution buffer pH 0.172 and the stop collect criterion 0.157, all significant and all positive, and the three significant interactions (load with elution pH, load with stop collect, and elution pH with stop collect) are the same three the screening block found. The conductivity of the load and wash buffer remains inactive, at a p-value of 0.14. No quadratic term is significant for any of the three responses, and the largest of them (the load quadratic in the aggregate model) reaches only 0.11. Over the ranges studied the surfaces are accordingly planes with an interaction twist, and there is no interior optimum and no edge of failure inside the region.

Table 5.7: Full quadratic coefficients for pool host cell protein, in model order.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	141.2	5.81	24.3	3.213e-12	***
A	52.55	3.96	13.27	6.207e-09	***
B	-49.72	3.96	-12.55	1.214e-08	***
C	4.142	3.96	1.046	0.3147	

Term	Coef.	Std. err.	t	p-value	Sig.
D	-2.13	3.96	-0.5377	0.5999	
AB	-16.25	4.201	-3.869	0.001937	**
AC	-4.265	4.201	-1.015	0.3285	
AD	-4.33	4.201	-1.031	0.3214	
BC	-13.99	4.201	-3.33	0.005424	**
BD	0.5473	4.201	0.1303	0.8983	
CD	-1.599	4.201	-0.3806	0.7097	
A^2	-16.36	10.46	-1.563	0.142	
B^2	12.95	10.46	1.237	0.2378	
C^2	2.307	10.46	0.2205	0.8289	
D^2	10.13	10.46	0.9681	0.3507	

The host cell protein coefficients in Table 5.7 reproduce the two dominant main effects and the protein load interaction with conductivity, and they add a second interaction that the screening block did not resolve. The coefficient for conductivity with elution buffer pH is -14.0 at a p-value of 0.0054, against a screening p-value of 0.33 for the same term (Table 14.4). The larger design and its lower residual variance are the reason the term is now visible, and the change should be read as better resolution and not as a different result. The effect is modest beside the two main effects, and it is reported here because it is real and because it slightly changes the shape of the acceptable region at the low conductivity edge.

Table 5.8: Analysis of variance for the pool aggregate response-surface model, with the residual partitioned into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	2.89	14	0.207	36.9	3.85e-08
Residual	0.0729	13	0.0056	nan	nan
Lack of fit	0.0673	10	0.00673	3.63	0.158
Pure error	0.00556	3	0.00185	nan	nan

Table 5.9: Analysis of variance for the pool host cell protein response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	104,028.0	14	7,430.6	26.3	3.07e-07
Residual	3,670.2	13	282.3	nan	nan
Lack of fit	2,626.7	10	262.7	0.755	0.679
Pure error	1,043.5	3	347.8	nan	nan

Neither model shows a significant lack of fit. For pool aggregate the lack-of-fit F test returns a p-value of 0.16 (Table 5.8), and for pool host cell protein 0.68 (Table 5.9). In both cases the unexplained variation is of the same size as the replicate variation, so the models do not miss structure that the centre points say is present. The lack-of-fit test for step yield is likewise not significant, at 0.96, and the full coefficient table for that response is in Appendix C.

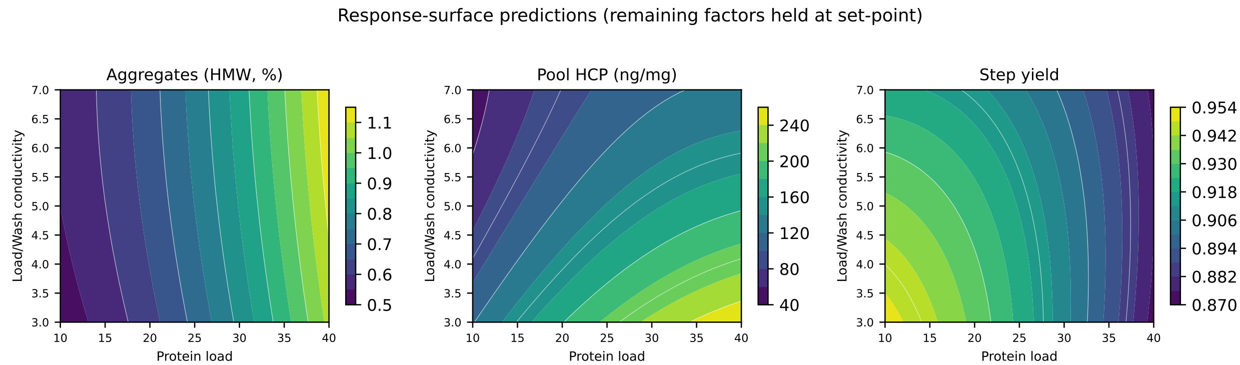

Figure 5.1: Response-surface predictions over protein load and load/wash conductivity, with elution buffer pH and the stop collect criterion held at their set-points.

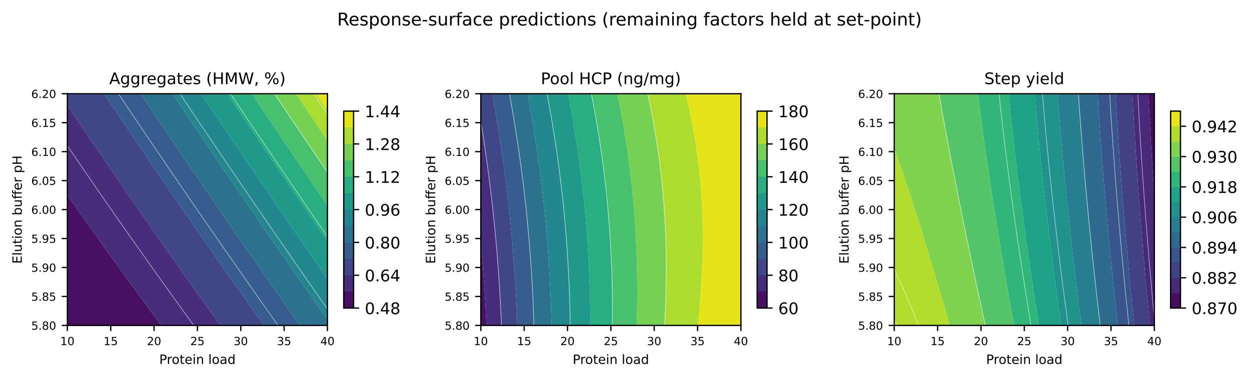

Figure 5.2: Response-surface predictions over protein load and elution buffer pH, with load/wash conductivity and the stop collect criterion held at their set-points.

The two planes that carry the argument are shown in Figure 5.1 and Figure 5.2. In the load and conductivity plane the host cell protein surface is strongly tilted and visibly twisted, which is the interaction seen in the coefficient table (Table 5.7). The aggregate surface in the same plane is almost flat along the conductivity axis, so the two attributes are governed by different parameters and can be controlled independently of one another. In the load and elution pH plane the aggregate surface tilts along both axes, and the twist between them is the load and elution pH interaction. The aggregate panels carry no contour at the drug substance limit, because no part of that surface comes close to it, and the host cell protein panels sit above the drug substance criterion throughout, for the reason set out in Section 6.

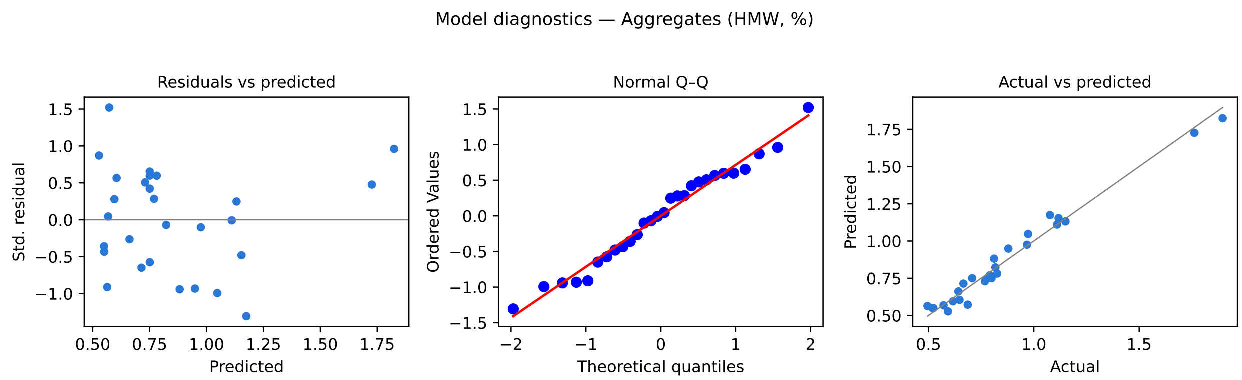

Figure 5.3: Model diagnostics for the pool aggregate response-surface model: standardized residuals against predicted values, normal quantile plot and actual against predicted.

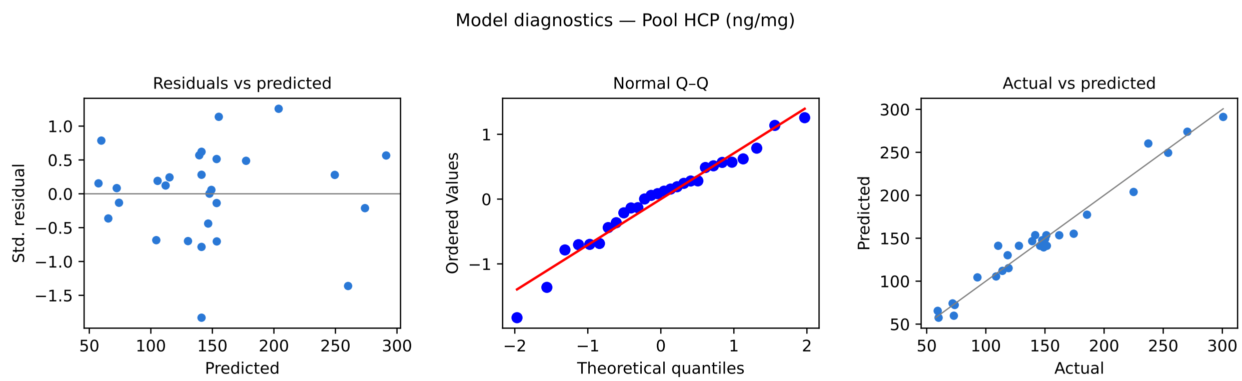

Figure 5.4: Model diagnostics for the pool host cell protein response-surface model.

The diagnostics in Figure 5.3 and Figure 5.4 support the two fits. The standardized residuals show no trend with the predicted value and no change of spread across the fitted range, so the constant-variance assumption holds. The normal quantile plots are close to linear, with no point far from the line. The actual against predicted plots lie on the identity line across the whole range of each response, including the extreme corners, which is the plot that matters when a model is used to predict at a corner of the region.

5.4 Mechanistic interpretation

The surfaces behave as the binding chemistry of the step predicts, and that agreement is what allows the models to be used at conditions between the ones actually run. Protein load acts on both governed attributes for the same underlying reason. As the load rises toward the working capacity of the resin, the column has less capacity available to resolve species that differ only slightly in their affinity for the resin. Aggregate, which binds more strongly than monomer because it carries a higher surface charge density, is displaced forward into the collected peak instead of remaining in the descending edge. Weakly bound host cell protein, which would otherwise be displaced by the antibody and removed in the wash, competes more successfully for the sites that remain. Step yield falls over the same range, from 0.938 at the low load edge to 0.876 at the high load edge, because product begins to appear in the flow-through as the column approaches capacity. All three responses accordingly move together with load, and the direction of each is the one prior knowledge expected (see Section 2).

The load and conductivity plane of the seeded process model is shown in Figure 5.5 for comparison with the fitted surfaces. It reproduces the same two features: a strongly curved host cell protein surface with its worst region at high load and low conductivity, and an aggregate surface that depends on load alone.

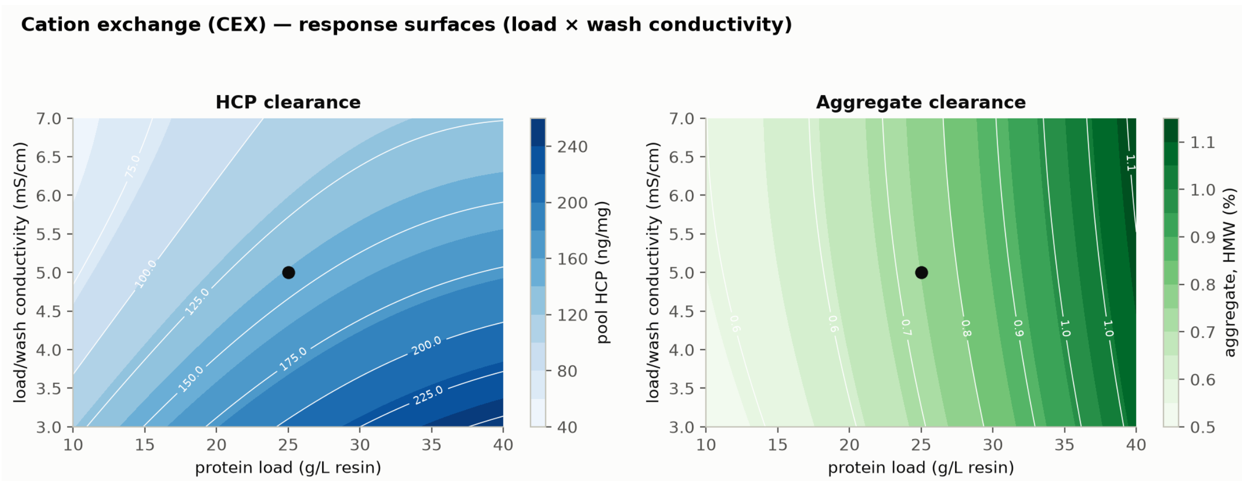

Figure 5.5: Process-model predictions for the cation exchange step over protein load and load/wash conductivity. The left panel is pool host cell protein and the right panel is pool aggregate; the marker is the step set-point.

The conductivity of the load and wash buffer acts on host cell protein alone, and the split is mechanistically clean. Raising the ionic strength weakens all electrostatic interactions with the resin, but it weakens the weak ones first. Host cell protein species retained by low avidity contacts are released and washed out, while the antibody, which binds through many contacts at once, stays on the column. Aggregate and monomer differ from each other in charge density and not in the number of weak contacts, so raising the conductivity moves the two species together and changes their separation very little. Conductivity is accordingly the parameter that governs pool host cell protein, and it has no detectable effect on pool aggregate (Table 5.6).

The interaction between protein load and conductivity has the same origin, expressed as a quantity. The amount of host cell protein presented to the column scales with the amount of material loaded, so the absolute quantity that a given wash strength can remove also scales with the load. Reading the fitted surface at the edges of the two ranges makes the size of the effect plain. At the high load edge, raising the conductivity from its low edge to its high edge lowers pool host cell protein by 132 ng/mg, while at the low load edge the same change in conductivity lowers it by only

67 ng/mg. The wash is worth roughly twice as much at the top of the load range as at the bottom, which is the practical content of the interaction and the reason the two parameters cannot be given independent univariate ranges.

Elution buffer pH and the stop collect criterion act on aggregate through the elution itself. Raising the elution pH lowers the net positive charge on the antibody and weakens its interaction with the resin, so the product elutes earlier and in a sharper front, and the aggregate that trails the monomer is pulled forward into the collected pool. Extending the stop collect criterion further down the descending edge collects more of the tail in which that aggregate sits. The two parameters act together, which is the significant interaction between them (term CD in Table 5.6), because the amount of aggregate available to be collected in the tail depends on how far forward the elution pH has already moved it. Both are therefore aggregate parameters, and neither has a significant effect on host cell protein, which has already been removed by the time the elution begins.

6 Design space

The design space for this step is the whole of the characterized four-dimensional region, because the governing attribute stays within its acceptance criterion at every point in that region. The claim rests on the aggregate response-surface model, and it is bounded in three ways that are stated at the end of this section.

Aggregate is the governing attribute, and the reason is structural. No unit operation downstream of cation exchange reduces aggregate, so the pool leaving this step determines the aggregate content of the drug substance and has to satisfy the drug substance criterion of 5 % high molecular weight species directly, without help from a later operation. The highest pool aggregate the model predicts anywhere inside the characterized ranges is 1.82 %, at the corner where protein load, load and wash conductivity, elution buffer pH and the stop collect criterion are all at their upper edges, which leaves 3.18 percentage points of margin to the criterion at the worst point of the region. The lowest predicted value, at the opposite corner, is 0.55 %, and within the normal operating ranges the worst corner falls to 1.27 %. The whole surface therefore sits in the lower part of the available range, and there is no edge of failure for aggregate inside the knowledge space.

Host cell protein does not bound the region either, but the argument for it is different and depends on the step that follows. The drug substance criterion of 100 ng/mg does not apply to the cation exchange pool, which is an in-process intermediate, and applying it here would require this step to deliver the whole host cell protein reduction on its own. What this step must deliver instead is a clearance factor, and a pool that anion exchange chromatography can finish. Anion exchange reduces pool host cell protein by a further factor of 6.2, from 117 ng/mg to 18.9 ng/mg in the nominal commercial-scale run (PCR-008). Dividing the drug substance criterion by that clearance factor gives 619 ng/mg as the cation exchange pool concentration from which the criterion can still be met. The worst corner of the characterized region for host cell protein is at high protein load with low conductivity, where the model predicts 291 ng/mg, a factor of 2.1 below that ceiling, and within the normal operating ranges the same corner falls further to 209 ng/mg. The cumulative host cell protein position across the whole train is consolidated in PCMR-001.

The two worst-case corners are not the same corner, and both were identified separately because the governing parameters differ. Aggregate is worst when every parameter is at its upper edge. Host cell protein is worst when protein load is high and the load and wash conductivity is low. The region has to satisfy both corners, and it does, with the margins given above. The normal operating ranges in Table 4.1 sit inside the characterized region on every axis, and the characterized region sits inside the knowledge space that the multivariate design and the univariate assessment together cover. This is the nesting that ICH Q8(R2) describes, and the design space claimed here is the multivariate region and not the set of univariate ranges (International Council for Harmonisation 2009).

Movement within the region is not a change in the regulatory sense, and it does not require prior approval (International Council for Harmonisation 2009). It remains subject to the change management system of the pharmaceutical quality system, so a planned move is assessed prospectively before it is made (International Council for Harmonisation 2008). Operation outside the region is a change and is handled through change control.

Three bounds apply to this claim, and none of them is removed by the data in this report. First, the region is defined over the ranges studied and says nothing about behaviour outside them, and the edges of the design space have not been confirmed at commercial scale. Second, the models predict mean responses, so the assurance that an individual batch meets the criterion comes from the capability analysis in Section 8 and not from the surfaces themselves. Third, all runs were executed on load material derived from a single low-pH viral inactivation pool, so the region describes the step for feed of that character and does not describe its response to a feed with a different aggregate or impurity burden. The aggregate and host cell protein content of the load is controlled by the upstream steps (PCR-003, PCR-005 and PCR-006) and is monitored as an input attribute, as described in Section 10.

7 Proven acceptable ranges

Every characterized parameter range is a proven acceptable range for aggregate. None is a proven acceptable range for host cell protein when that attribute is judged against the drug substance criterion, and the reason is the cross-step dependency described in Section 6 and not any shortcoming of the step.

The acceptance basis is the drug substance specification in both cases, taken from Table 2.1. Neither response is a viral clearance response, so no back-calculation from a cumulative requirement is involved, and the criterion applied to each is the one the drug substance itself must meet. Two analyses were run for each combination of attribute and parameter. The first holds the other three parameters at their set-points (coded zero) and scans the parameter of interest across its characterization range. The second varies the other three within their normal operating ranges by Monte-Carlo simulation of the fitted response-surface model, and asks over what part of the range the 95 % predictive interval of the attribute stays inside the criterion. The second analysis is the stricter of the two, because it requires the parameter to remain acceptable while the rest of the operating point moves around it. Both are given in Table 7.1.

Table 7.1: Proven acceptable ranges for each governed attribute and each multivariate parameter, at the set-point and with the remaining parameters varying within their normal operating ranges.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Protein load	10–40	g/L resin	10–40	10–40
Aggregates (HMW, %)	Load/Wash conductivity	3–7	mS/cm	3–7	3–7
Aggregates (HMW, %)	Elution buffer pH	5.8–6.2	pH	5.8–6.2	5.8–6.2
Aggregates (HMW, %)	Elution stop collect	0.5–1.5	OD desc.	0.5–1.5	0.5–1.5
Pool HCP (ng/mg)	Protein load	10–40	g/L resin	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Load/Wash conductivity	3–7	mS/cm	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Elution buffer pH	5.8–6.2	pH	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Elution stop collect	0.5–1.5	OD desc.	none (set-point breaches)	none (set-point breaches)

For aggregate the two analyses give the same answer for all four parameters, and that answer is the full characterization range in each case. Protein load is acceptable across its whole studied range (10 to 40 g/L resin), and so are the load and wash conductivity, the elution buffer pH and the stop collect criterion. The coincidence of the two columns is the informative part, because it says that each parameter remains acceptable not only when the others are held at their set-points, but also when they vary within their normal operating ranges, which is the condition the commercial process will actually meet. No parameter needs a proven acceptable range narrower than the range that was characterized.

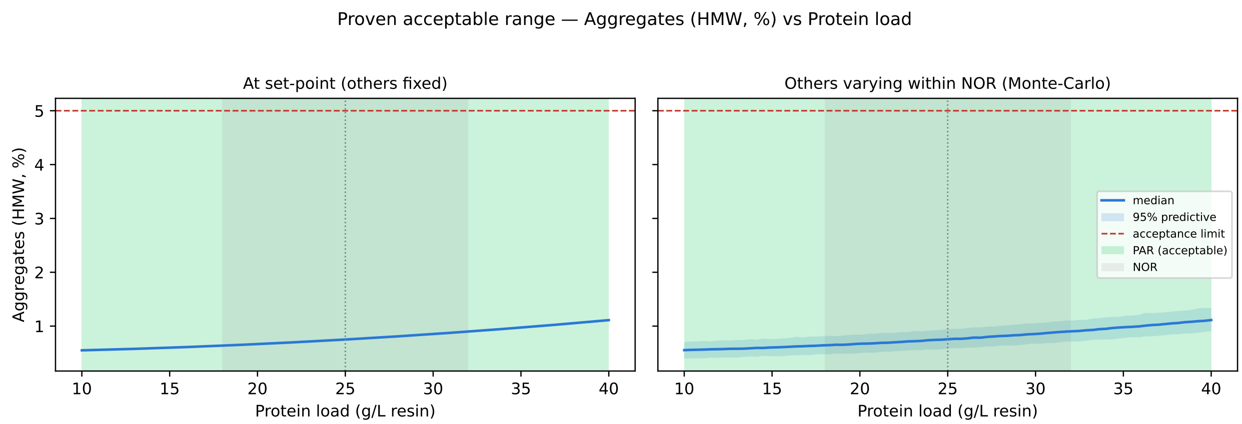

Figure 7.1: Proven acceptable range for pool aggregate against protein load. The left panel holds the other parameters at their set-points; the right panel varies them within their normal operating ranges by Monte-Carlo simulation. The acceptable region is shaded green.

Figure 7.1 shows the analysis for aggregate against protein load, which is the parameter with the largest effect on that attribute (Table 5.3). The predicted response rises across the load range and stays far below the acceptance limit throughout, in both panels. The shaded acceptable region covers the whole axis, and the upper edge of the predictive interval in the right-hand panel remains below the limit even at the top of the load range, where the combination of a heavily loaded column and normal operating range variation in the other three parameters produces the highest aggregate the analysis considers.

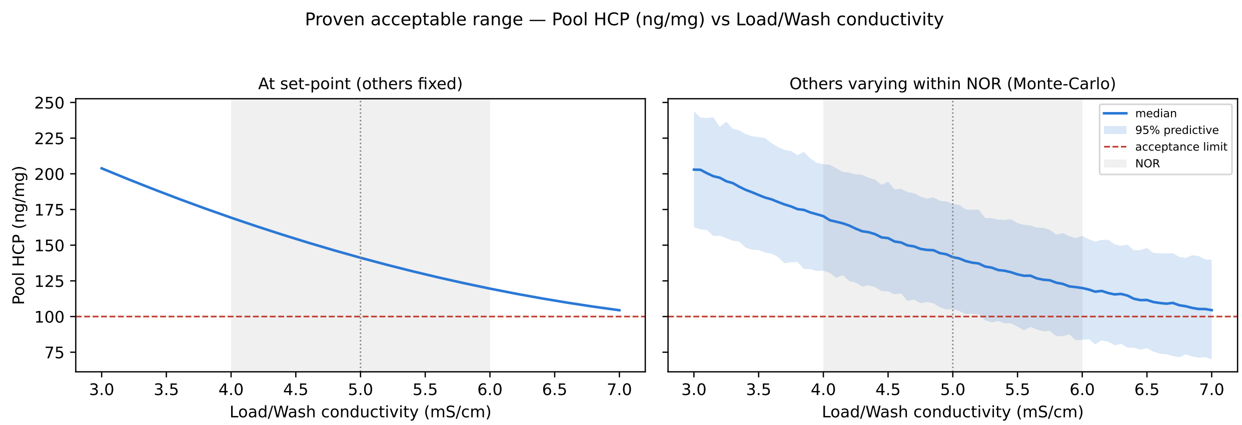

Figure 7.2: Pool host cell protein against load/wash conductivity, analysed against the drug substance criterion. No region is acceptable against that criterion, because the criterion applies to the drug substance and not to this intermediate.

For host cell protein the analysis returns no acceptable range for any parameter, and Figure 7.2 shows why. The predicted pool concentration lies above the drug substance criterion across the whole conductivity range, including at the set-point, so no contiguous acceptable interval exists around the set-point for the analysis to report. This is the correct result for an intermediate that is not the drug substance, and it should not be read as a failed acceptance criterion. Applying a drug substance criterion to the cation exchange pool would require this step to deliver the whole host cell protein reduction on its own, which is neither how the process is designed nor what is claimed for it.

The quantity this step is judged on for host cell protein is its clearance factor, which is 78-fold in the nominal commercial-scale run, together with the step-level ceiling derived in Section 6. Judged that way, every parameter range is acceptable. The highest concentration the model predicts anywhere in the characterized region is 291 ng/mg, against a ceiling of 619 ng/mg. Since the worst point of the whole region satisfies the ceiling, every sub-range of every parameter satisfies it too, and no narrower proven acceptable range is needed for any of the four. This conclusion is conditional on the anion exchange clearance factor holding, which is a claim of PCR-008 and not of this report, and the cumulative outcome is reported in PCMR-001.

The proven acceptable ranges above and the design space in Section 6 are different claims, and they are stated separately here so that neither is read as the other. The proven acceptable ranges are univariate, made either at the set-point or under normal operating range variation of the other parameters. The design space is the multivariate region, and it is the region that governs. Where every proven acceptable range spans the full characterization range, as it does here, the binding constraint on the operating region is the edge of the characterized space itself and not any univariate limit. Both claims are bounded by the ranges studied and by the fitted response-surface models, and neither extends to conditions outside the characterization ranges given in Table 4.1.

8 Process capability and robustness

The four attributes this step governs all meet their drug substance criteria at commercial scale with wide margins. Capability was estimated by Monte-Carlo simulation of 2,000 batches, with every process parameter in the train drawn within its normal operating range, and is reported as a one-sided Cpk against the relevant criterion in Table 8.1.

Table 8.1: Commercial-scale capability for the quality attributes governed by the cation exchange step.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.11
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.81
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2785.30

Aggregate has a Cpk of 16.1, on a simulated mean of 0.75 % high molecular weight species against a limit of 5 %. The margin between the mean and the limit is 4.25 percentage points, or more than 48 standard deviations of the simulated batch distribution (the simulated standard deviation is 0.088 %). This capability belongs to the cation exchange step alone, because no other step in the train changes the attribute.

Host cell protein has the tightest capability of the four, at a Cpk of 6.1 on a mean of 15.2 ng/mg against a limit of 100 ng/mg. That figure is a whole-train result and the credit for it is shared across three steps. Protein A chromatography removes the largest single share and is reported in PCR-005, this step removes a further 78-fold in the nominal commercial-scale run, and anion exchange chromatography removes a further 6.2-fold and is reported in PCR-008. Residual DNA and leached Protein A have Cpk values of 10.8 and 2785. Both are dominated by the Protein A step (PCR-005) and by anion exchange chromatography (PCR-008), and the contribution credited to cation exchange is 1.5 and 1.0 log reduction respectively.

The tightest capability in the drug substance process as a whole is not one of these four. It is 1.51, and it belongs to the cumulative viral clearance attributes (XMuLV and MVM), which this step does not claim and which are consolidated in PCMR-001. A wide capability at this step therefore says nothing about the tightest constraint on the process, and the two should not be read together.

Two bounds apply to the capability figures. They are computed with every parameter inside its normal operating range, so they describe routine operation and not operation at the edge of the design space defined in Section 6. They also rest on the scale-down model being representative of the commercial step, which is argued in the materials and methods section and confirmed at the centre point only.

9 Parameter classification

Each parameter was classified from its demonstrated effect on the governed attributes and from the reliability with which it can be held inside the operating region, following SOP-4001. The classifications are given in the last column of Table 4.1 and are reasoned individually below.

- Protein load is a well-controlled critical process parameter (WC-CPP). It has the largest effect on pool aggregate, which is the attribute this step carries, and the second largest on pool host cell protein. It is classified as well controlled and not as critical because it is set by a calculation from the measured load concentration and the packed resin volume, both of which are determined before the cycle begins, so the risk of an unnoticed excursion outside the design space is low.
- Load and wash conductivity is a well-controlled critical process parameter. It governs pool host cell protein and has no detectable effect on pool aggregate ($p = 0.14$). It is measured in line throughout the load and the wash, so an excursion is detected while the cycle is running and can be corrected before the elution.
- Elution buffer pH is a well-controlled critical process parameter. It has the second largest effect on pool aggregate and it interacts with protein load. The buffer is prepared to a specification and released before use, and the pH is verified in line at the column inlet, which is why the parameter is well controlled despite its narrow range (5.8 to 6.2).
- The stop collect criterion is a well-controlled critical process parameter. It affects pool aggregate through how much of the descending edge enters the pool, and it is enforced by the ultraviolet detector and the collection logic, so it cannot drift between cycles.
- Elution flow rate is a general process parameter (GPP). The univariate assessment found no effect on any governed attribute across its characterization range, and prior knowledge places its action on cycle time. It is retained in the knowledge space as evidence that the separation is insensitive to residence time over that range.

Three factors that showed no effect on a given response are retained in the same way. Load and wash conductivity has no effect on pool aggregate, and elution buffer pH and the stop collect criterion have no effect on pool host cell protein. Those null results simplify the control strategy, because they mean the aggregate and the host cell protein controls act through different parameters and do not have to be traded against one another. They are also evidence of robustness over the ranges screened, and they are recorded as such rather than dropped.

No parameter at this step is a critical process parameter, and none is a key process parameter. All 4 quality-linked parameters are well controlled, and the one remaining parameter (the elution flow rate) is a general process parameter. Across the drug substance process the classification counts are CPP 1, GPP 6, KPP 10, WC-CPP 20, and the single critical process parameter belongs to the low-pH viral inactivation step (PCR-006), where the pH range is narrow and the consequence of an excursion is a loss of viral inactivation.

10 Contribution to the control strategy

The cation exchange step contributes four elements to the control strategy for A-Mab drug substance, and it is the sole source of one of them. Aggregate control is the element it holds alone. Because no downstream operation reduces aggregate, the operating region defined in Section 6 and the pool release testing described below are together the entire assurance that the drug substance meets its aggregate criterion.

The first element is parametric control. The four well-controlled critical process parameters are held within the normal operating ranges given in Table 4.1, under SOP-2010, and classification is applied under SOP-4001. Movement within the wider design space is permitted under the change management provisions described in Section 6.

The second element is in-process monitoring of the pool. Pool aggregate is measured by size-exclusion chromatography (AMV-3011) on every batch, and pool host cell protein by immunoassay (AMV-3012). Residual DNA and leached Protein A are monitored under AMV-3014 and AMV-3016 at the frequency set in the control strategy. The aggregate result is the in-process control that confirms, batch by batch, the assurance the design space provides prospectively, and it is the only such confirmation the attribute receives, since no later step can correct an out-of-trend result.

The third element is control of the material entering the step. The aggregate content of the load is set upstream, at the production bioreactor (PCR-003) and through the low-pH hold (PCR-006), and it was held constant across this study. It is monitored as an input attribute so that a load outside the characterized range is detected before the cycle begins. The host cell protein content of the load is set by Protein A chromatography (PCR-005) and is monitored on the same basis.

The fourth element is life-cycle control of the resin. Packing quality, cycle number, sanitization and storage are governed by SOP-2008. The characterization runs reported here were executed on resin within its qualified cycle range, so the operating region is claimed for resin in that condition and not beyond it, and resin performance at the end of its qualified life is addressed by the life-cycle programme and not by this study.

Three things this step does not do should be stated alongside those it does. It makes no viral clearance claim, and the cumulative clearance for A-Mab rests on low-pH inactivation, anion exchange chromatography and virus filtration (PCR-006, PCR-008 and PCR-009). It does not deliver the host cell protein, DNA or leached Protein A criteria on its own, and the credit for those is shared as described in Section 8. It does not modify charge variants or glycan variants, and no claim is made for either, because those attributes are formed at the production bioreactor (PCR-003) and were not measured across this study. All four contributions and the three non-claims are consolidated in PCMR-001.

11 Discussion

The step is now understood in the terms the control strategy needs. Two attributes vary measurably with the operating conditions, and each is governed by a different set of parameters. Aggregate follows protein load, elution buffer pH and the stop collect criterion, which are the parameters that determine where the collected pool sits relative to the aggregate-enriched tail of the elution peak. Host cell protein follows protein load and the conductivity of the load and wash buffer, which are the parameters that determine what is removed before the elution begins. Protein load is the only parameter the two sets share, and the separation of the rest is a useful property, because it means the two controls do not compete for the same setting.

The most consequential single finding is that the aggregate limit is not approached anywhere inside the characterized region. The highest predicted value is 1.82 % against a limit of 5 %, and the commercial-scale capability is a Cpk of 16.1 (Table 8.1). That is a comfortable position for the attribute on which this step has sole responsibility, and it should not be read as a claim that the step is robust without qualification. The result holds over the ranges studied, for load material of the character used in this study, and for the mean response predicted by the fitted models.

Three limitations qualify the conclusions, and each is managed forward rather than resolved here. The first is the feed. Every run used load derived from a single low-pH viral inactivation pool, so the study characterizes the response of the step to its operating parameters and not to variation in the aggregate or impurity content of its load. That variation is real, and it is controlled at source and monitored as an input attribute, as set out in Section 10. The second is scale, because the scale-down model was qualified against the commercial step at the centre point only, and the design space edges have not been confirmed at commercial scale. The third is the resin life cycle, which is outside the scope of this study and is covered by the programme under SOP-2008.

The step yield model deserves a separate comment, because its adequacy is materially weaker than that of the two attribute models. Its predicted coefficient of determination is 0.20, which is too low for the model to be used for prediction at a stated condition. What the yield data do support is the direction and the approximate magnitude of the load effect, and the finding that no other parameter affects yield measurably (Table 14.5). Yield is a process performance response and not a quality attribute, so this limitation constrains process economics and not the quality claim.

The 2 deviations recorded during execution are described in Section 13. Neither changed the fitted effects, neither altered a classification, and both were dispositioned as retained. No parameter at this step required the most stringent control class. All 4 quality-linked parameters are held either by instruments that read during the cycle or by calculations completed before it starts, which is why each is classified as well controlled and not as critical.

12 Conclusions

The cation exchange step is well characterized, and the operating region it needs for commercial manufacture is defined. Over the four multivariate parameters the response-surface models describe pool aggregate and pool host cell protein with a predicted coefficient of determination of 0.82 and 0.83, and neither shows a significant lack of fit. The design space is the whole characterized region, because the governing attribute stays within its criterion at every corner of it.

Aggregate meets its drug substance criterion at commercial scale with a Cpk of 16.1, and this step carries that attribute alone. Host cell protein, residual DNA and leached Protein A meet their criteria with the credit shared across Protein A chromatography, this step and anion exchange chromatography. The tightest capability among the four attributes governed here is host cell protein, at a Cpk of 6.1.

All 4 multivariate parameters are classified as well-controlled critical process parameters, and the elution flow rate is a general process parameter. Every proven acceptable range spans the full characterization range of its parameter, at the set-point and under normal operating range variation of the others. The campaign recorded 2 deviations, both of which were investigated and dispositioned as retained, without effect on any fitted model or any classification. No viral clearance is claimed for this step, and no claim is made for modification of charge or glycan variants. These conclusions hold over the ranges studied, for the scale-down model as qualified, and for load material of the character used here. They roll up into PCMR-001.

13 Deviations from the plan

The campaign recorded 2 deviations from PCP-007 during execution. Each was investigated to a root cause, assessed for impact on the study conclusions, and dispositioned. Both were dispositioned as retained, which means the affected run stayed in the analysis rather than being excluded or repeated. Neither was found during execution: one was raised in the post-execution documentation review and the other in the equipment history review, so both were assessed against data that had already been collected. The register is given in Table 13.1, and each event is described below.

Table 13.1: Register of deviations recorded during execution of PCP-007.

Deviation	Summary	Detected during	Disposition
DEV-007-01	Expired Tris load/wash buffer lot used in screening run 6	post-execution documentation review	retained
DEV-007-02	Column inlet temperature excursion during RSM run 14	equipment history review	retained

13.1 DEV-007-01 — expired buffer lot used in a screening run

A buffer lot that had passed its expiry date was used to prepare one screening run. Lot LOT-BUF-2287 of the Tris load and wash buffer carried an expiry date of 2026-02-11 and was used in screening run 6, which is a factorial run at low protein load and high load and wash conductivity. The lot was not checked against its expiry date at the point of preparation, which is the recorded root cause. The event was found in the post-execution documentation review, after the screening block had been executed and analysed.

Retained material from the lot was re-tested under SOP-2103, which authorises retained-sample testing when an expiry deviation is raised. The re-test returned a pH of 5.02 and a conductivity of 5.06 mS/cm. Neither result indicates that the buffer had changed in a way that would alter the binding or the wash behaviour of the step.

The impact on the study was assessed against the fitted screening models. The largest standardized residual of run 6 across the three responses is 0.68, so the run agrees with the fitted models to well within one residual standard deviation. Refitting each screening model with the run excluded changes the significance of 0 model terms at the 5 % level, which is the check that matters here, because the screening block exists to identify the active factors and nothing in that identification depends on the run. The run was accordingly retained in the analysis. The preparation record for the step has since been amended to require an expiry check at buffer preparation.

13.2 DEV-007-02 — column inlet temperature excursion in a response-surface run

The column inlet temperature was not held at its set-point during one response-surface run. During run 14 the inlet temperature rose to 26.4 °C and remained above the chamber set-point for 38 minutes. The cause was a controller set-point drift on the temperature-controlled chromatography chamber (EQ-CHR-118), and the event was found in the equipment history review. The chamber was inside its calibration interval at the time of execution, with calibration due on 2026-05-30, so the drift was a control fault and not a measurement fault.

An elevated temperature during chromatography can promote aggregation, so the assessment focused on the aggregate response of the affected run. 3 verification runs were executed at the run 14 condition with the chamber under control, and their pools were assayed for aggregate by the verification-qualified size-exclusion method AMV-3221 (relative standard deviation 3.8 %, limit of quantitation 0.2 % high molecular weight species). The relative 95 % confidence half-width of the verification set mean is 9.44 %, and that figure is the resolution of the comparison that follows.

The measured aggregate for run 14 differs from the value predicted by the response-surface model by 3.1 %, which is inside that resolution. The excursion accordingly produced no aggregate signal the study could detect, and the run was retained. Refitting each response-surface model with the run excluded changes the significance of 0 model terms at the 5 % level, so no coefficient, no operating region boundary and no classification depends on it. The controller has been recalibrated and an alarm limit added to the chamber control loop.

14 Appendices A–D

The appendices carry the complete design matrices, the full effect and coefficient tables for every response, and the analytical method performance summary. Tables shown in the results sections are subsets of these.

14.1 Appendix A — Screening design matrix and responses

The coded screening design and its measured responses are given in Table 14.1. Factor codes are those of Table 4.2, and the coded levels are minus one, zero and plus one.

Table 14.1: Screening design matrix in coded factors, with the measured responses.

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	factorial	-1	-1	-1	-1	0.5	136	0.936
2	factorial	-1	-1	-1	1	0.604	114	0.962
3	factorial	-1	-1	1	-1	0.548	135	0.94
4	factorial	-1	-1	1	1	0.818	138	0.973
5	factorial	-1	1	-1	-1	0.452	66.2	0.934
6	factorial	-1	1	-1	1	0.564	61.3	0.951
7	factorial	-1	1	1	-1	0.579	48.8	0.946
8	factorial	-1	1	1	1	0.752	67.6	0.934
9	factorial	1	-1	-1	-1	0.792	428	0.883
10	factorial	1	-1	-1	1	1.1	247	0.888
11	factorial	1	-1	1	-1	1.14	301	0.923
12	factorial	1	-1	1	1	1.79	240	0.908

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
13	factorial	1	1	-1	-1	0.807	105	0.895
14	factorial	1	1	-1	1	1.04	113	0.893
15	factorial	1	1	1	-1	1.12	115	0.884
16	factorial	1	1	1	1	1.83	149	0.886
17	center	0	0	0	0	0.833	153	0.929
18	center	0	0	0	0	0.829	183	0.935
19	center	0	0	0	0	0.692	129	0.91

14.2 Appendix B — Response-surface design matrix and responses

The coded response-surface design and its measured responses are given in Table 14.2. Axial runs are placed on the faces of the design cube, so every coded setting is minus one, zero or plus one.

Table 14.2: Response-surface design matrix in coded factors, with the measured responses.

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	factorial	-1	-1	-1	-1	0.518	109	0.98
2	factorial	-1	-1	-1	1	0.593	114	0.955
3	factorial	-1	-1	1	-1	0.572	151	0.973
4	factorial	-1	-1	1	1	0.818	142	0.942
5	factorial	-1	1	-1	-1	0.495	59.3	0.921
6	factorial	-1	1	-1	1	0.616	71.9	0.946
7	factorial	-1	1	1	-1	0.686	60.1	0.932
8	factorial	-1	1	1	1	0.812	73	0.925
9	factorial	1	-1	-1	-1	0.768	237	0.87

Run	Type	A	B	C	D	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
10	factorial	1	-1	-1	1	0.973	254	0.878
11	factorial	1	-1	1	-1	1.15	301	0.9
12	factorial	1	-1	1	1	1.76	270	0.89
13	factorial	1	1	-1	-1	0.826	174	0.902
14	factorial	1	1	-1	1	1.12	139	0.903
15	factorial	1	1	1	-1	1.08	118	0.891
16	factorial	1	1	1	1	1.9	119	0.844
17	axial	-1	0	0	0	0.523	73.7	0.935
18	axial	1	0	0	0	1.11	186	0.879
19	axial	0	-1	0	0	0.666	225	0.92
20	axial	0	1	0	0	0.791	92.9	0.909
21	axial	0	0	-1	0	0.647	149	0.921
22	axial	0	0	1	0	0.879	148	0.914
23	axial	0	0	0	-1	0.642	162	0.935
24	axial	0	0	0	1	0.967	150	0.922
25	center	0	0	0	0	0.796	146	0.881
26	center	0	0	0	0	0.708	110	0.952
27	center	0	0	0	0	0.782	128	0.921
28	center	0	0	0	0	0.8	152	0.917

14.3 Appendix C — Full effect and coefficient tables

Table 14.3 to Table 14.5 give the complete screening effect tables for all three responses, and Table 14.6 gives the response-surface coefficient table for step yield. The response-surface coefficient tables for pool aggregate and pool host cell protein are in the results section.

Table 14.3: Complete screening effect table for pool aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.5987	0.2993	0.024	12.47	1.595e-06	***
C	0.3401	0.1701	0.024	7.087	0.0001033	***
D	0.319	0.1595	0.024	6.648	0.0001612	***
AC	0.1958	0.0979	0.024	4.08	0.003536	**
AD	0.1544	0.07718	0.024	3.216	0.0123	*

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
CD	0.1313	0.06565	0.024	2.736	0.02561	*
B	-0.01847	-0.009235	0.024	-0.3848	0.7104	
BC	0.01515	0.007577	0.024	0.3157	0.7603	
BD	-0.01313	-0.006566	0.024	-0.2736	0.7913	
AB	0.01205	0.006024	0.024	0.251	0.8081	

Table 14.4: Complete screening effect table for pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-126.6	-63.29	8.85	-7.151	9.701e-05	***
A	116.4	58.22	8.85	6.579	0.0001732	***
AB	-56.75	-28.37	8.85	-3.206	0.0125	*
BD	39.6	19.8	8.85	2.237	0.05568	
D	-25.84	-12.92	8.85	-1.46	0.1824	
AD	-24.59	-12.3	8.85	-1.389	0.2022	
CD	24.44	12.22	8.85	1.381	0.2048	
BC	18.37	9.183	8.85	1.038	0.3298	
AC	-12.51	-6.254	8.85	-0.7067	0.4998	
C	-9.449	-4.724	8.85	-0.5338	0.608	

Table 14.5: Complete screening effect table for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.05184	-0.02592	0.002867	-9.041	1.792e-05	***
BC	-0.01213	-0.006067	0.002867	-2.116	0.06722	
B	-0.01106	-0.005529	0.002867	-1.929	0.08991	
AD	-0.009137	-0.004568	0.002867	-1.594	0.1497	
D	0.006831	0.003416	0.002867	1.191	0.2676	
C	0.00666	0.00333	0.002867	1.162	0.2789	
BD	-0.005664	-0.002832	0.002867	-0.9879	0.3521	
CD	-0.004726	-0.002363	0.002867	-0.8242	0.4337	
AC	0.004009	0.002005	0.002867	0.6992	0.5042	
AB	3.376e-05	1.688e-05	0.002867	0.005889	0.9954	

Table 14.6: Response-surface coefficient table for step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.9175	0.006511	140.9	4.361e-22	***
A	-0.03073	0.004439	-6.924	1.046e-05	***
B	-0.007574	0.004439	-1.706	0.1117	
C	-0.003583	0.004439	-0.8072	0.4341	

Term	Coef.	Std. err.	t	p-value	Sig.
D	-0.005482	0.004439	-1.235	0.2386	
AB	0.008028	0.004708	1.705	0.1119	
AC	0.0002686	0.004708	0.05705	0.9554	
AD	-0.0006968	0.004708	-0.148	0.8846	
BC	-0.006465	0.004708	-1.373	0.1929	
BD	0.001848	0.004708	0.3925	0.701	
CD	-0.006584	0.004708	-1.399	0.1853	
A^2	-0.01034	0.01172	-0.8821	0.3937	
B^2	-0.002783	0.01172	-0.2373	0.8161	
C^2	0.0005746	0.01172	0.04901	0.9617	
D^2	0.01083	0.01172	0.9238	0.3724	

Table 14.7: Analysis of variance for the step yield response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.0217	14	0.00155	4.38	0.00573
Residual	0.00461	13	0.000355	nan	nan
Lack of fit	0.0021	10	0.00021	0.251	0.958
Pure error	0.00251	3	0.000836	nan	nan

14.4 Appendix D — Analytical methods summary

Table 14.8 lists the validated methods applied to the responses of this study with their reported performance characteristics. AMV-3221 is the verification-qualified size-exclusion method used for the deviation assessment in Section 13.

Table 14.8: Analytical methods applied to the cation exchange characterization, with reported performance characteristics.

Method	Analytical procedure	Precision (%RSD)	Limit of quantitation	Accuracy (%)
AMV-3011	Size-Variants (SEC-HPLC)	qualified	qualified	qualified
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95
AMV-3014	Residual DNA (qPCR)	qualified	qualified	qualified
AMV-3016	Leached Protein A by ELISA	6.5	0.25 ppm	97.4
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8	0.2 % HMW	99.1

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